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2026 SECONDARY 2 COMMUNICATION SCIENCE			
S/NO.: 2	TOPIC: Signals	UNIT: Practical Electricity	NOTES 2

Knowledge	Cognitive Skills or Practical Skills
2.1 Power	Cognitive Skills (a) Explain what is meant by power, relate it to an output of an electrical system, stating its SI unit. (b) Apply the relationships $P = V.I$ and $E = V.I.t$
2.2 Cost of electricity	(c) Calculate the cost of using electrical appliances, using kilowatt-hour as a unit of electrical energy consumption. (d) Show an awareness of ways to conserve electrical energy in our households.
2.3 Hazards	(e) State how changes made to an electrical system can cause some electrical hazards.
2.4 Safety measures	(f) State some precautionary measures and technologies used to ensure the safe use of electricity in the home (e.g. circuit breaker, earth leakage)
2.5 Energy Sources & Energy Security	(g) Show an appreciation of the uses of various sources of energy (e.g., fossil fuels, solar, hydro- electric, wind energy, geothermal energy, biofuels and nuclear energy) and their impact on the environment (h) Recognise that man uses man-made technologies (e.g. photovoltaic cells and turbines) to harness energy from natural resources. (i) Discuss the importance for Singapore, which has no natural resources of her own, to conserve energy and how this could help reduce contribution to global warming.
Essential Questions	1. (F) What are the practical applications of electricity? 2. (C) How can electricity affect us in our daily lives?

2.1 Deriving the concept of Power and Energy in electricity**Potential difference**

- Potential difference is also defined as the work done per unit charge.
- It is expressed as

$$V = W / Q = E / Q$$

where V = potential difference
 Q = charge
 W = work done
 E = energy transferred

Current

- Current is the rate of flow of charges.
- It is expressed as

$$I = Q / t$$

where I = current
 Q = charge
 t = time

Power

- Power is defined as the rate of work done or rate of energy conversion.
- Power is expressed as

$$P = E / t = W / t$$

where P = Power
 W = Work done
 E = Energy transferred
 t = time

- The SI unit of power is the Watt (**W**).

Electrical Power

- If we combine the three equations above,
- $P = E / t = (Q \cdot V) / t = V \cdot (Q / t) = V \cdot I$

$$P = V \cdot I$$

where P = Power
 V = Potential difference
 I = Current

- The unit of power is the Watt, **W**.

Other forms of equation for Electrical Power

- If we combine with the equation $V = I.R$,
- $P = V.I = I.R.I = I^2R = (V/R)^2.R = V^2/R$

$$P = I^2R = V^2/R$$

where P = Power
 V = Potential difference
 I = Current
 R = Resistance

Example 1

A filament lamp which is rated at 60 W, 240 V, is connected to a 240 V power supply.

(a) Find the current flowing through the lamp when it is working normally.

$$\begin{aligned}\text{amps} &= \text{watts/voltage} \\ &= 60\text{W} / 240\text{V} \\ &= 0.25\text{A}\end{aligned}$$

(b) Find the resistance of the lamp.

$$\begin{aligned}I &= V/R \\ R &= V/I \\ R &= 240\text{V} / 0.25 \\ &= 960\Omega\end{aligned}$$

Example 2

An electric heater has the following label “1000 W, 240 V”.

(a) Explain what it means.

When connected to 240 V, the heater will transfer 1000 joules of energy to a different store per second

(b) Calculate the amount of electrical energy consumed when the heater is used for 20 minutes.

$$\begin{aligned}20\text{mins} &= 1200\text{s} \\ 1000\text{V} &= E/1200\text{s} \\ E &= 1200\text{s (time)} \times 1000\text{V (power)} \\ &= 1\,200\,000\end{aligned}$$

2.2 Calculating the cost of electricity

- Each month, every household is billed for the total of amount of electricity they consume throughout the month; in Singapore, it is Singapore Power (under Public Utilities Board) that bill consumers for the electricity they use.
 - An electricity meter (see image on the right) is used to monitor the electricity consumption of a household.
 - The difference between meter's reading for the current month and the previous month gives the electrical energy used for the current month.
 - The cost of electricity consumed is calculated based on the energy unit **kilowatt-hour (kWh)**.
 - One kilowatt-hour is the amount of **energy** used by a 1kW device in 1 hour.
 - The kilowatt-hour is a unit of Energy and **NOT** a unit of Power.
 - 1 kWh = 1 kW x 1h = 1000 W x 3600 s = 3 600 000 J = 3.6 MJ**
 - The cost of electricity is usually charged at **\$0.30 per kWh**.
- *Tariff as of Jul 2023**



Electricity Meter

Example 3

A 1.5 kW electric coil heats water for 2.0 hours.

(a) Calculate the electrical energy consumed by the coil in kWh.

$$\begin{aligned}
 \text{Energy consumed} &= Pt \\
 &= 1.5 \text{ kW} \times 2.0\text{h} \\
 &= 3.0 \text{ kWh}
 \end{aligned}$$

(b) Express the electrical energy in Joules, J.

$$\begin{aligned}
 3.0 \text{ kW} \times 1000 \times 1\text{h} \times 60 \times 60 &= 3,600 \text{ MJ} \\
 &\text{or } 11 \text{ MJ}
 \end{aligned}$$

Example 4

The following figure shows an electricity bill that was dated 28 Dec 2009.

CURRENT MONTH CHARGES		Usage	Rate (\$)	Amount (\$)	Total (\$)
Electricity Services					
Reading taken on 11-12-2009 : 16780					
Electricity		68 kWh	0.2169		

Calculate the total cost of electricity that was due.

$$\text{Cost} = 68\text{kWh} \times \$0.2169/\text{kWh} = \$14.75$$

Activity 1: Energy Consumption at Home

Aim: To introduce students to the concept of electricity pricing (kWh cost) at home.

You are required to find out typical consumption of different home appliances and equivalent dollar value of energy usage using the table below. The main items at home are listed for your reference. Assume that 1 kWh cost \$0.23.

SN	Device	Number of units	*Power rating / W	Daily usage / h	Total energy used / kWh	Total cost / \$
1	Air conditioner	5	3000	8h	120	27.60
2	Water Heater	3	3000	1h 30mins	13.5	3.105
3	Refrigerator	1	200	24h	4.8	1.104
4	Internet Router	1	20	24h	0.48	0.1104
5	Phone Charger	3	10	5h	0.15	0.0345
					Daily total	31.95
* Please check against the back of your appliance at home.					Monthly total	990.57

Based on your individual home usage pattern, can you recommend the appropriate measures (changing habits, use of alternative appliances) to reduce energy usage?

Try to use it lesser and make sure that electrical appliances are turned off

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Keep a daily log for another week where there is intentional responsible use of electricity.

SN	Device	Number of units	*Power rating / W	Daily usage / h	Total energy used / kWh	Total cost / \$
1	Air conditioner	5	3000	10h	150	34.50
2	Water Heater	3	3000	1h	9	2.07
3	Refrigerator	1	200	24h	4.8	1.104
4	Internet Router	1	20	24h	0.48	0.1104
5	Phone Charger	1	10	24h	0.24	0.0552
					Daily total	37.84
* Please check against the back of your appliance at home.					Monthly total	1173.03

Based on your individual home usage pattern for the 2 days, reflect on the most significant energy saving method.

Turn off the switch if not in use

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2.3 Hazards of Electricity

Many of us come into close contact with electricity every day at home and in school. Yet, many are not aware of the hazards and dangers involved in the use of electricity.



Overloading of power sockets, water and frayed cables are common hazards.

Watch the following YouTube video of Electrical Safety: <http://bit.ly/2uENWca> (7 min approx.) and then answer the following questions.

(a) State 4 common hazards of using electricity at home and / or in school.

1. Leaving the appliances turned on
2. Open wires
3. Water near electricity wires
4. Overloading causes the main wire to be very hot and may even cause a fire.

(b) Explain how the hazards stated in (a) are dangerous.

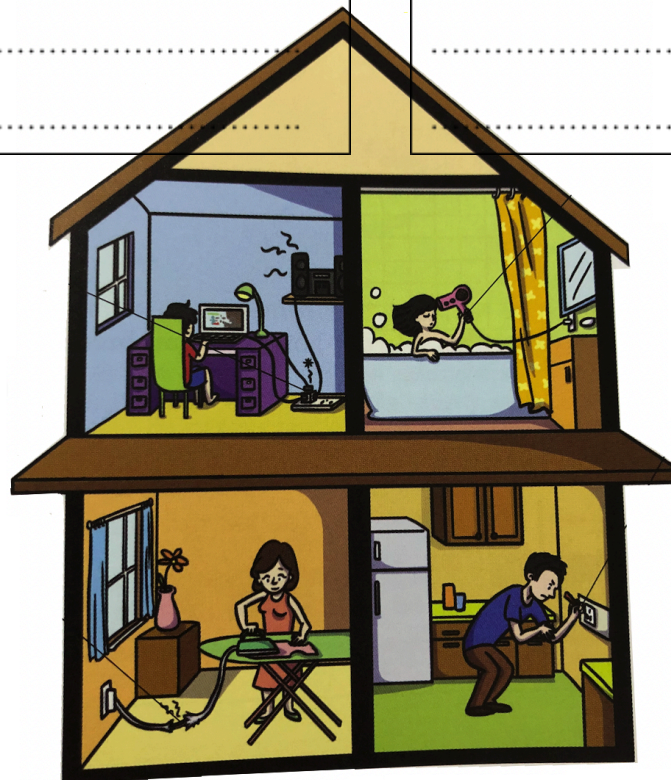
1. If there is anything flammable, it can burn
-
2. If there is anything flammable, it can burn
-
3. People who contact with the water can be electrocuted
-
4.
-

Activity 2: Safety precautions in using electricity

Based on the cartoons below, describe the precautions to take when using electricity.

Overloading causes the main wire to be very hot and may even cause a fire.

If the hairdryer drops into the water, the person bathing will get electrocuted



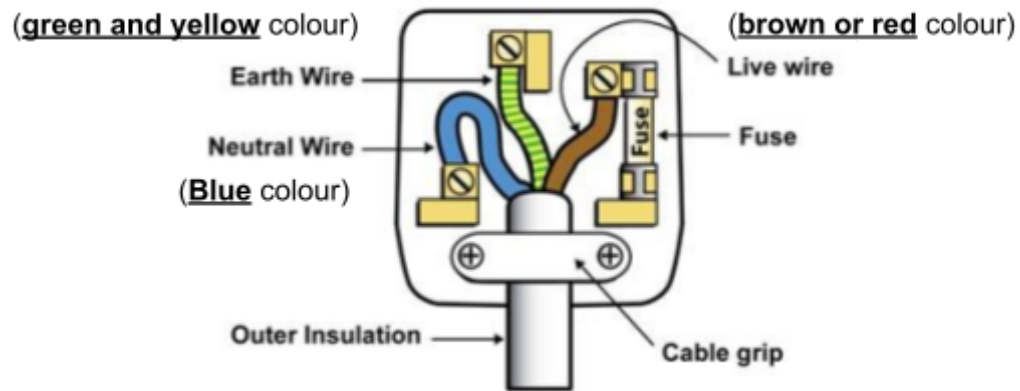
The wire is exposed, someone touching it will get electrocuted

Put something metal in a plug and the man can get electrocuted

2.4. Safety features in home circuits

Despite the hazards, there are many common safety features that incorporated into a typical **electrical wiring system** to protect users from danger.

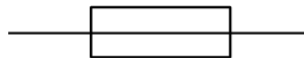
2.4.1 The British Standard 3-Pin Plug



- There are **three types** of wire inside the 3-pin plug.
- The live wire is connected to a **high voltage** and delivers **an electric current** to the appliance. The **fuse** and **switch** are fitted to the live wire.
- The neutral wire **completes the circuit** by providing a returning path to the supply. It is usually set at **zero volts**.
- The earth wire is a low-resistance wire connected to the **metal casing** of appliances. This wire is to allow excess current to flow to the earth if there is a leakage of current.
- <https://www.youtube.com/watch?v=At3aRDH2fjs>

2.4.2 Fuses & Fuse Rating

The figure on the left shows the circuit symbol for a fuse and a photo of a fuse is shown on the right.



- Fuses are safety devices placed in electric circuits to **break** the circuit when there is an **excessive flow** of electric current.
- Fuses should have a current rating just **slightly higher** than the current an electrical appliance will use under normal conditions.
- A fuse should be connected to the **live** wire so that the electric appliance will not become charged.
- Typical fuse ratings are **1A, 2A, 3A, 5A, 10A and 13A**.

Example 5

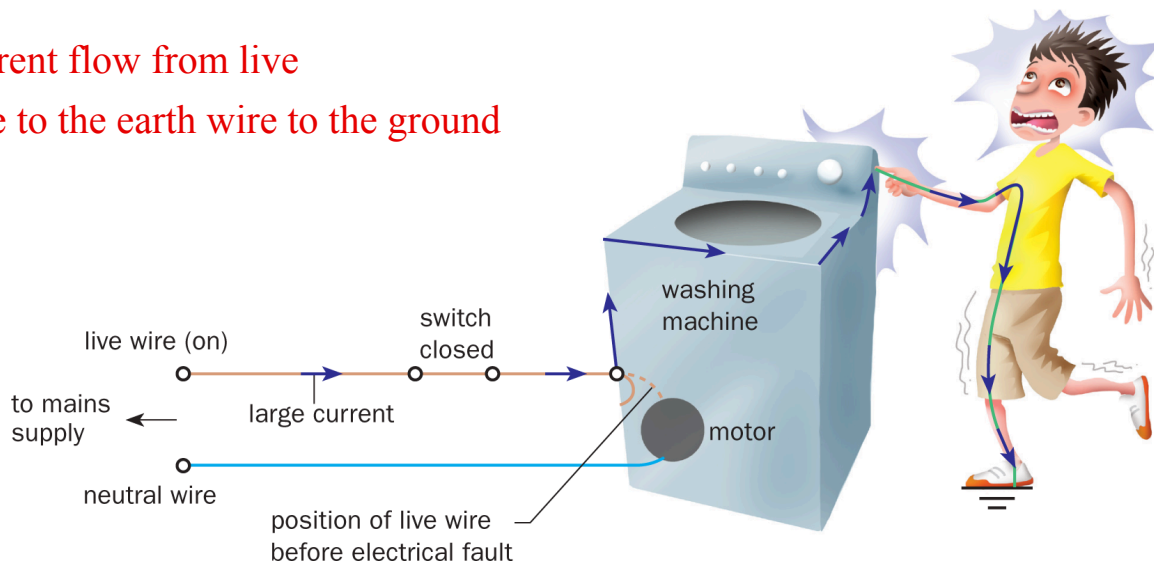
Suggest a suitable fuse rating for a 3000 W, 240 V electric water heater.

13A

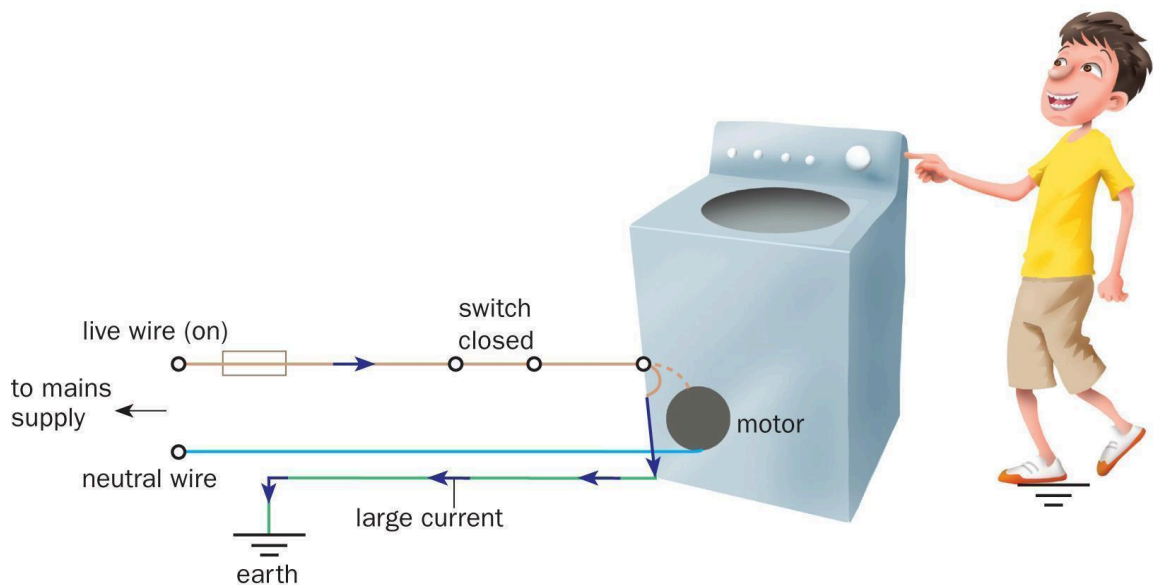
2.4.3 Earth wire on metal casing

- Electrical current can leak from the live wire to the metal casing of an appliance when there is an electrical fault.
- In the absence of earth wire:

Current flow from live wire to the earth wire to the ground



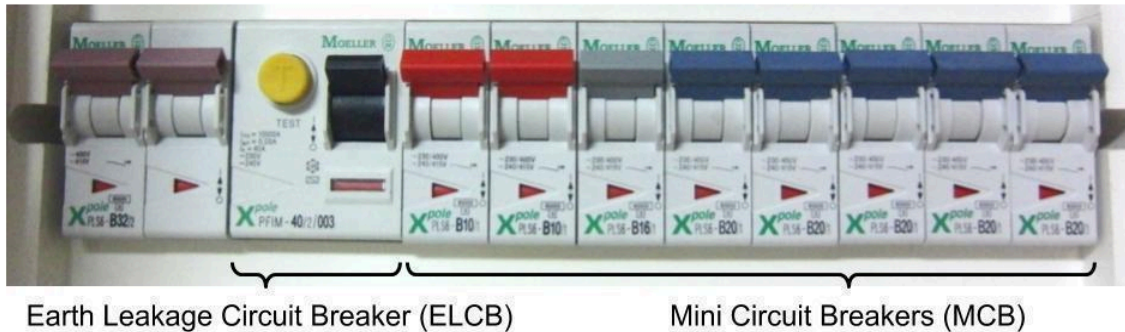
- In the presence of an earth wire:



- The earth wire can **reduce** electrical shocks when there is a **short-circuit** between the live wire to the casing.

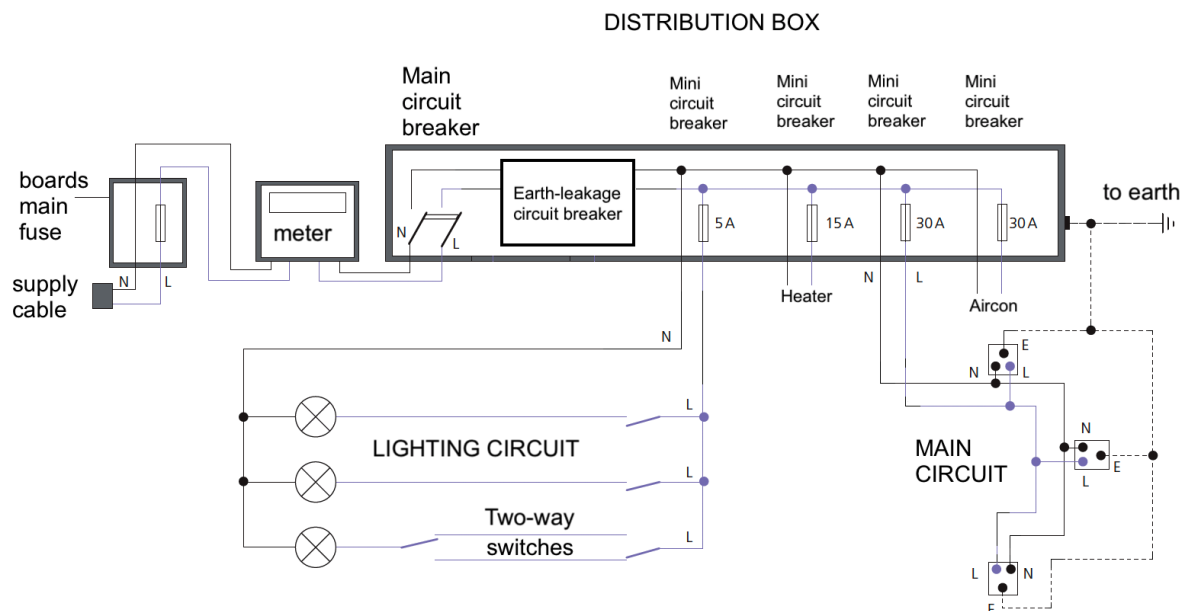
2.4.4 Circuit Breakers

Circuit breakers are safety devices that cut off the electrical supply in a circuit when there is an overflow of electric current. You should be able to find one in most households within Singapore (see figure below).



- The ELCB trips if there is an unbalanced current (**>30 mA**) in the live and neutral wires.
- The MCB trips if there is a current that is above the rated value (**e.g. 6A, 20A**).

The ELCB or RCCB (Residual Current Circuit Breaker) and MCB (Mini-Circuit Breakers) are found in the Distribution Box (DB) in the household wiring system.



Activity 3: Tripping the MCB & ELCB

The picture to the right is a mockup of a household panel. There are altogether

- (a) 4x lighting unit
- (b) 1x air-conditioning unit
- (c) 4x 3-pin sockets
- (d) 1x water heater unit

Each set of units is connected to a mini circuit breaker (MCB) and an earth leakage circuit breaker (ELCB).



- (a) Connect devices to the 3-pin plugs. Increase the number of devices until the MCB trips.

What could be the cause?

Overloading. There are too many devices drawing current from the mains

- (b) When the ELCB trips, what could be the cause?

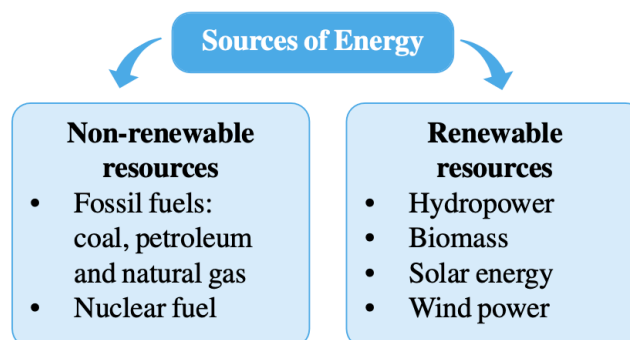
There is a current leakage through the earth wire. The unbalanced current causes the ELCB to trip.

What should you do?

If you can identify the device, get it repaired by a licensed electrician

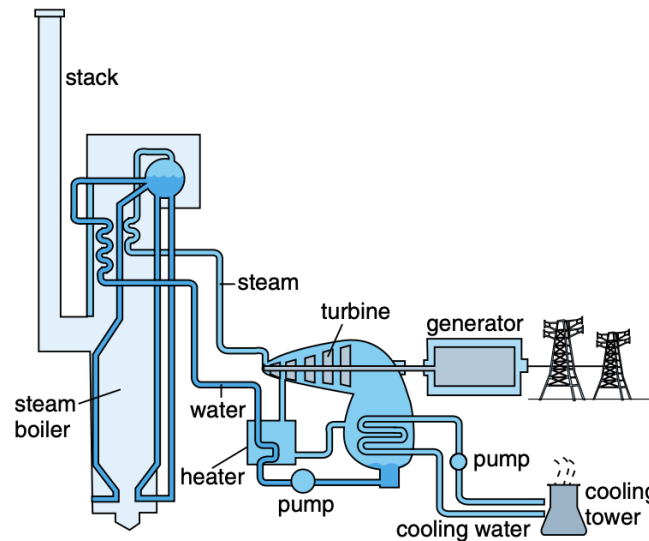
2.5 Sources of Energy**2.5.1 Renewable and Non-renewable sources of energy**

1. The consumption of electrical energy is very large. It has become a necessity in our daily life.
2. Electrical energy must be generated as it does not occur naturally (except during lightning). It is generated from other sources of energy.
3. Sources of energy can be divided into two classifications; **non-renewable resources** and **renewable resources**.



2.5.2 Generation and Environmental Impacts

2.5.2.1 Fossil Fuels

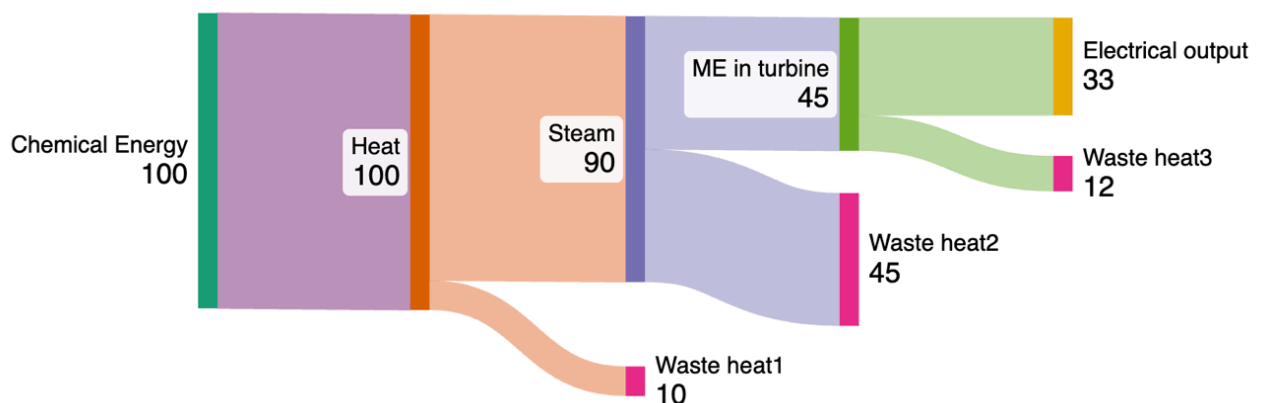


Thermal power station

Production process

- Fuels such as coal, petroleum and natural gas are burned in a furnace to produce heat used to boil water in a boiler. Steam at high pressure is formed.
- The high-pressure steam drives a turbine which produces the rotational motion needed to run the generators.
- Not all of the energy produced in the turbine is converted into electrical energy. Only about 33% of the heat energy input is converted into electrical energy output.

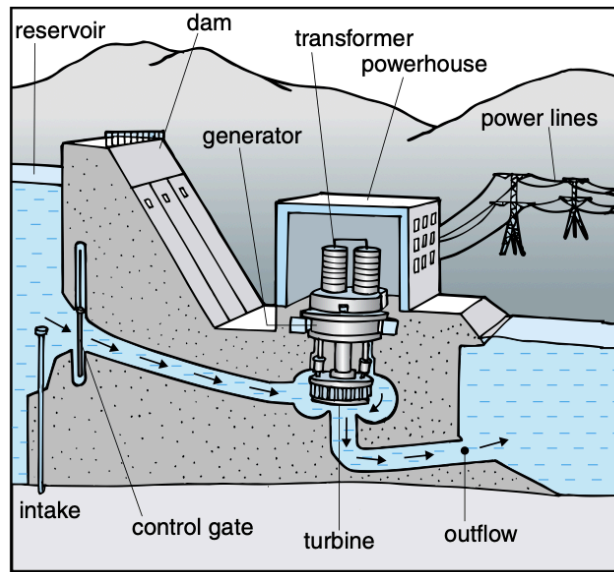
Energy Transfer



Effects on the environment

- Smoke, dust and toxic gases such as sulfur dioxide and nitrogen oxides, which pollute the air, are produced.
- Excessive carbon dioxide in the atmosphere causes the greenhouse effect which gives rise to higher temperatures in the environment (global warming).
- Acid rain due to the solution of sulfur dioxide and nitrogen oxides in water has an adverse effect on the equilibrium of the earth's ecosystem.

2.5.2.2 Hydroelectricity

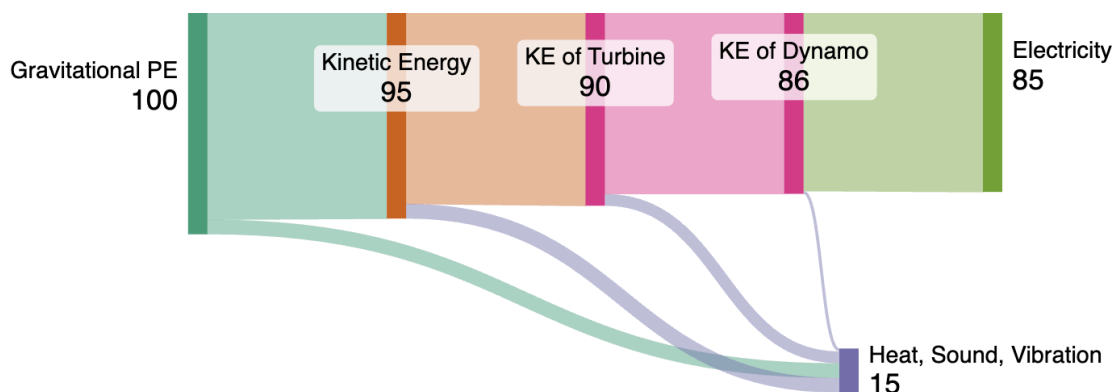


Hydropower plant

Production process

- A hydropower station uses the kinetic energy of falling water to produce electrical energy.
- Water is stored behind a massive wall of concrete which retains water in a valley to a high level— called a dam.
- The water is then released and flows downhill through turbines which drive electrical generators. Gravitational potential energy is changed to kinetic energy, and then to electrical energy.

Energy Transfers



Effects on the environment

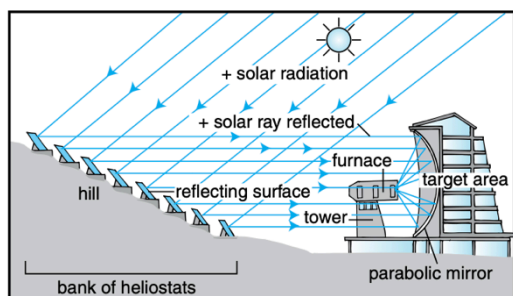
The construction of a dam disturbs the equilibrium of the ecosystem in the environment because a large area of forest land has to be destroyed to build the dam. Also, the flow of a river is affected. Fish ecosystems will be affected as the fish cannot travel along the rivers blocked by the dam.

A huge water dam, when it is no longer in use, may cause flooding if the dam collapses due to poor maintenance and ageing.

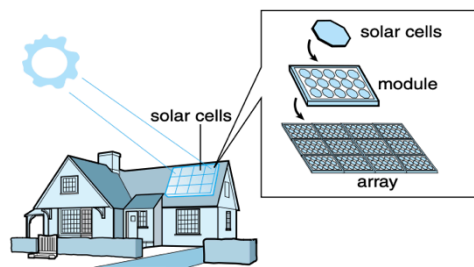
2.5.2.3 Solar Energy

Solar energy is energy from the Sun. Solar energy is converted into electrical energy by two methods: in a solar furnace and in solar cells (photocells).

Solar Furnace



Solar Cells (Photocells)



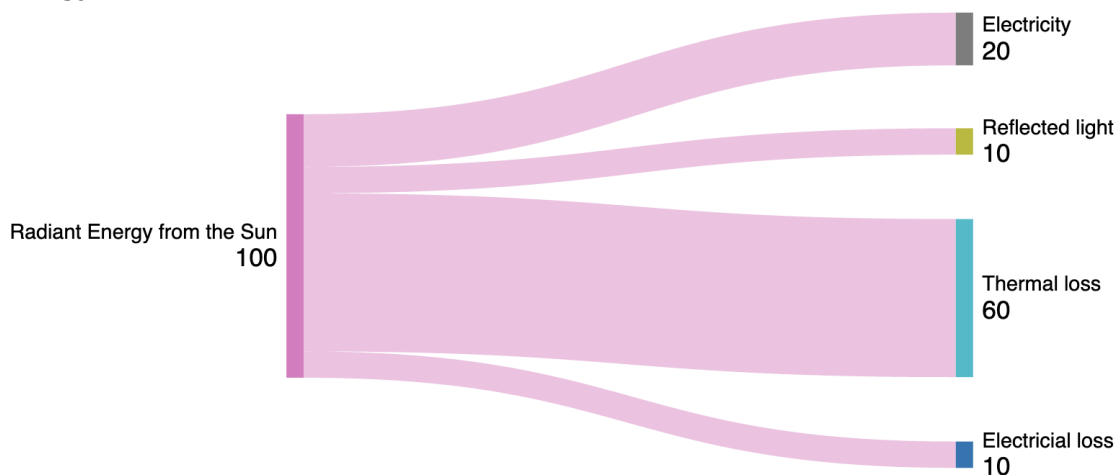
Production process (Solar furnace)

A solar thermal power station has a large parabolic mirror to focus the Sun's rays to a solar furnace. The heat energy is used to heat water to produce steam whose kinetic energy is used to drive the turbine of an electrical generator.

Production process (Solar cells)

- A solar cell consists of a semiconductor material such as silicon which is able to convert light energy into electrical energy.
- Light energy releases electrons from the silicon atoms. The free electrons produced flow in the material and a current is produced.
- Solar cells are largely used in wrist watches, lamps, calculators and water heaters for domestic use. More complex photocells are used in satellites.

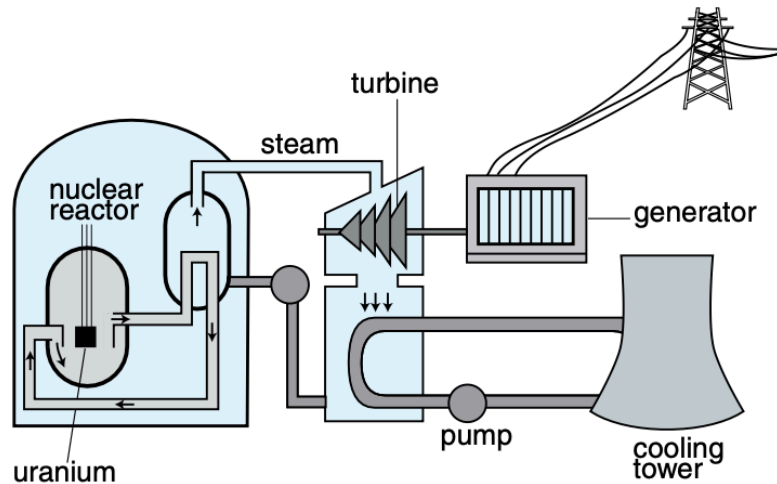
Energy Transfers (in solar cells)



Effects on the environment

- The facility for collecting the Sun's rays to converge to a solar furnace or onto solar panels requires a large area.
- Mirrors and solar panels, when it has reached its expired stage, can cause pollution to the environment if not disposed off properly.
- Batteries, which are an essential storage for electricity produced by solar panels, can cause environmental pollution if not disposed of properly.

2.5.2.4 Nuclear Energy

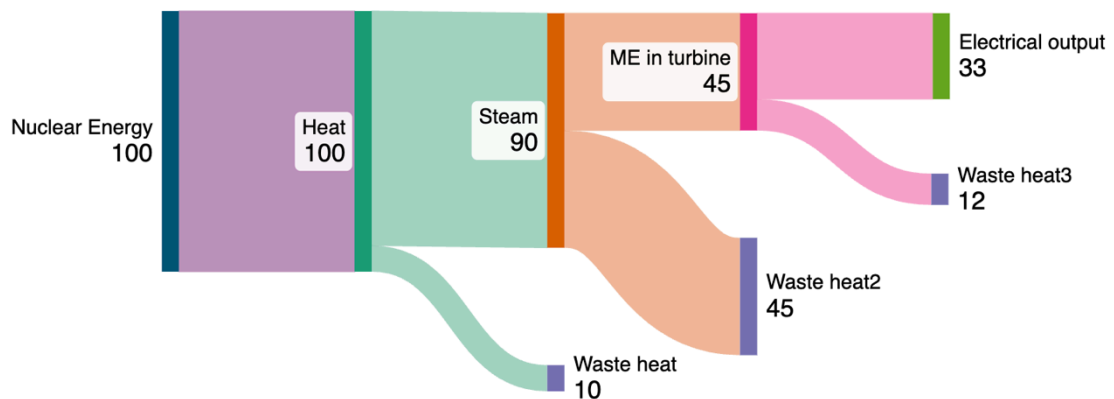


Nuclear Energy Plant

Production process

- In a nuclear reactor, the fission of a uranium atom is carried out at a controlled rate.
- The heat energy released in the fission process is used to produce high-pressure steam. The high-pressure steam is used to drive a turbine that would then rotate a generator as in a thermal power station.

Energy Transfers

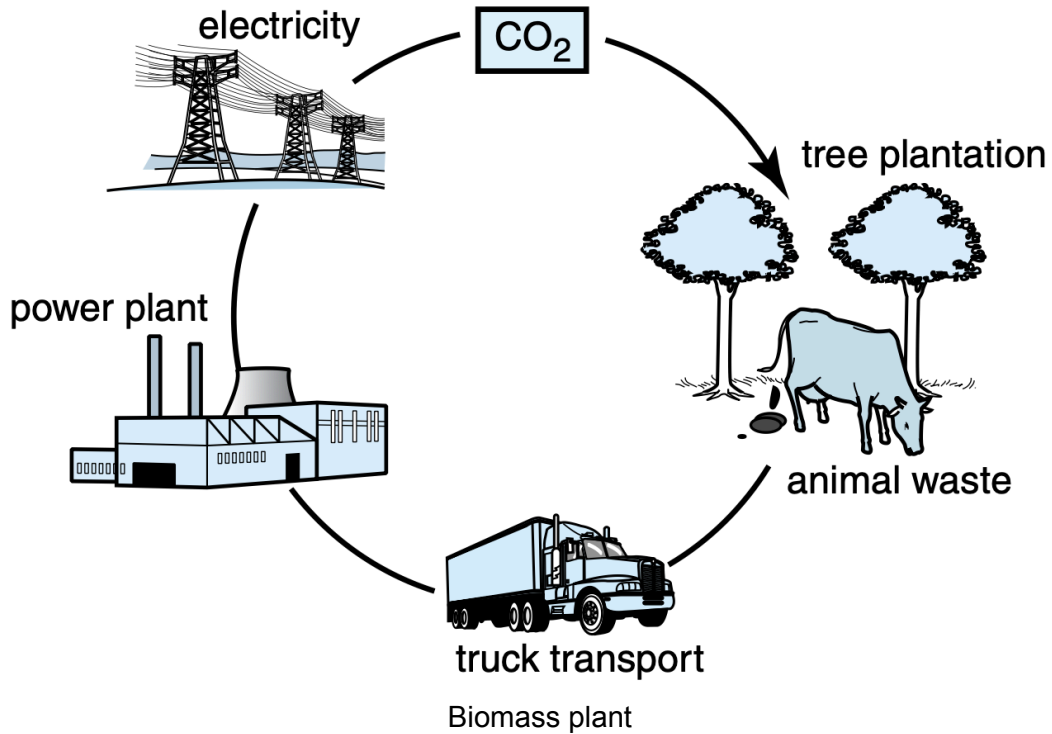


Effects on the environment

- The fuel used in a nuclear reactor cannot be easily and safely stored. Handling of the radioactive waste requires careful management.
- Any occurrences of leakage in a nuclear power station will contaminate the atmosphere with dangerous radioactive radiation, causing death to humans and animals. Radioactive waste released to the atmosphere is spread by the wind and is a threat to health.
- The cost of constructing a nuclear reactor is very high because of the technology which needs to be acquired and the necessary human resources to manage the facility.

2.5.2.5 Energy from Biomass

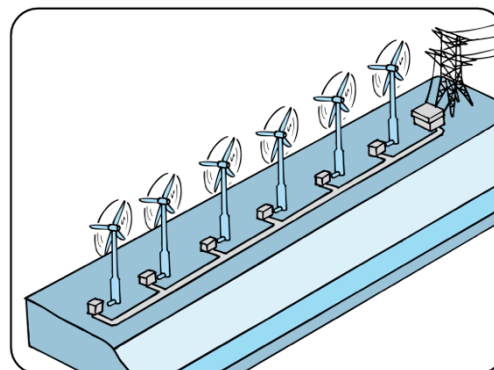
Biomass is organic material from plants or waste matter of animals.

**Production process**

- (a) Energy obtained from plant materials such as wood or waste matter from animals is called biomass energy.
- (b) The decaying process by the action of bacteria on plant and animal waste produces biogas such as methane which can be used as fuel for the production of electrical energy.
- (c) Burning of biomass produces heat energy to generate electrical energy.

Effects on the environment

- (a) Biomass must be kept in designated areas to avoid the spread of foul smells from affecting the health of residents.
- (b) Management of a biomass facility must be suitably qualified in science and technology.

2.5.2.6 Wind Power*Windmill***Production process**

- (a) Wind turbines are used by farmers in windy areas.
- (b) A wind turbine consists of two or three blades connected to the rotor of an electrical generator.
- (c) The wind turns the blades, which spin a shaft, which connects to a generator and creates electricity.
- (d) A wind turbine with a blade of 50 m can produce electrical power up to 1 MW.

Effects on the environment

- (a) The construction of wind turbines requires a large area near the peaks of mountains or along the beaches. Forests have to be cleared for the wind farms.
- (b) The rotation of the turbines produces noise.

Activity 6: Research on other energy sources

Although 6 energy sources are described above, there are many more sources of energy that are being developed. Carry out research on any one of the following and describe its method of production and its effects on the environment. Your research should cover the following areas: **Production, Energy Transfer, Advantages of the Energy Source, and Effects on the Environmental.**

Source of Energy	Index number of student
Nuclear Fusion	1, 7, 13, 19, 25
Nuclear Energy using Thorium	2, 8, 14, 20
Geothermal Energy	3, 9, 15, 21
Tidal Power	4, 10, 16, 22
Wave Power	5, 11, 17, 23
Radio Thermal Generator	6, 12, 18, 24

Activity 7: Appreciating energy security issues

Electricity is very important in our daily lives, as evidenced by the great inconvenience when it is not available. As Singapore has no natural resources, it depends largely on imports of natural resources such as natural gas or crude oil to produce electricity.

Do you know that there were major blackouts that took place in Singapore previously?

In August 2002, Indonesia suddenly cut off the natural gas pipe that was linked to Singapore. This resulted in 7 out of 9 gas-fired generator plants tripping. As a result of the disruption, many residential and industrial areas suffered blackouts.

In June 2004, this time the gas valve at West Natuna, Indonesia, was disrupted again. The impact of this disruption caused 3 major power stations in Singapore to trip and caused 30% of Singapore to be in darkness.

In December 2006, the natural gas supply from Malaysia was cut off, and this disruption led to the power supply cutoff to businesses and residents in the Northeast, East and Central of Singapore.

Source: Ng, W. H. (2013). *Singapore, the energy economy: from the first refinery to the end of cheap oil, 1960-2010*.

Discussion question: What should we do to ensure that Singapore do not meet with such an unpredictable crisis in the future?

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Activity 8: Importance of saving electricity – Open Energy Market

Starting 1 April 2018, EMA will commence the soft launch of Open Electricity Market (previously referred to as Full Retail Competition), where households and businesses in Jurong can choose to buy electricity from a retailer with a price plan that best meets their needs. This will provide consumers with choice and flexibility in their electricity purchases. An EMA spokesman said: "Consumers will no longer be restricted to buying electricity from SP Group at the regulated tariff, and can stand to benefit from more competitive pricing and innovative offers from electricity retailers."

Task: Your mum tasked you to evaluate the different vendors that can provide the best rest for a fixed price plan over a period of **12 months**. You can consider other important aspects for your family, such as **cost-effectiveness**, **reliability**, and **environmental friendliness**.

Conduct short research on the company by comparing the price plans: <http://compare.openelectricitymarket.sg>.

Suggestions to parents:

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THE END